

Near-Field Communications for 6G: Fundamentals, Challenges, Potentials, and Future Directions

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Abstract—Extremely large antenna array (ELAA) is a common feature of several key candidate technologies for sixth-generation mobile networks (6G), such as ultra-massive multiple-input-multiple-output (UM-MIMO), cell-free massive MIMO, reconfigurable intelligent surface (RIS), and terahertz communications. Since the number of antennas is very large for ELAA, the electromagnetic radiation field needs to be modeled by near-field spherical waves, which is opposed to the conventional planar-wave-based radiation model of 5G massive MIMO. As a result, near-field communications will become essential in 6G wireless networks. In this article, we systematically investigate the emerging near-field communication techniques. Firstly, we present the fundamentals of near-field communications and the metric to determine the near-field ranges in typical communication scenarios. Then, we investigate recent studies specific to near-field communications by classifying them into two categories, i.e., techniques addressing the challenges and those exploiting the potentials in near-field regions. Their principles, recent progress, pros and cons are discussed. More importantly, several open problems and future research directions for near-field communications are pointed out. We believe that this article would inspire more innovations for this important research topic of near-field communications for 6G.

Index Terms—6G, ELAA, near-field communications, spherical wavefront.

I. INTRODUCTION

Sixth-generation mobile networks (6G) are promising to empower emerging applications, such as holographic video, digital replica, etc. For fulfilling these visions, tremendous research efforts have been endeavored to develop new wireless technologies to meet the key performance indicators (KPI) of 6G, which are much superior to those of 5G [1]. For instance, thanks to the enormous spatial multiplexing and beamforming gain, ultra-massive multiple-input-multiple-output (UM-MIMO) and cell-free massive MIMO (CF-MIMO) are promising to achieve a 10-fold increase in the spectral efficiency for 6G [1]. Besides, by dynamically manipulating the wireless environment through thousands of antennas, reconfigurable intelligent surface (RIS) brings new possibilities for capacity and coverage enhancement [2]. Moreover, millimeter-wave (mmWave) and terahertz (THz) UM-MIMO can offer abundant spectral resources for supporting $100\times$ peak data rate

improvement (e.g., Tbps) in 6G mobile communications [3].

Despite being suitable for different application scenarios with various KPIs, all the above technologies, including UM-MIMO, CF-MIMO, RIS, and THz communications, share a common feature: They all usually require a very large number of antennas to fulfill their expected performance, i.e., extremely large antenna arrays (ELAA) are essential for these different promising candidate technologies for 6G.

Compared with massive MIMO, the key technology for 5G, ELAA for 6G not only means a sharp increase in the number of antennas but also results in a fundamental change of the electromagnetic (EM) characteristics. The EM radiation field can generally be divided into far-field and near-field regions. The boundary between these two regions is determined by the Rayleigh distance, which is also termed as the Fraunhofer distance [4]. Rayleigh distance is proportional to the product of the square of array aperture and carrier frequency [4]. Outside the Rayleigh distance, it is the far-field region, where the EM field can be approximately modeled by *planar waves*. Within the Rayleigh distance, the near-field propagation becomes dominant, where the EM field has to be accurately modeled by *spherical waves*.

Since the number of antennas is not very large in 1G-5G wireless networks, the Rayleigh distance of up to several meters is negligible. Thus, existing 1G-5G communications are mainly developed from far-field communication theories and techniques. However, with the significant increase of the antenna number and carrier frequency in future 6G systems, the near-field region of ELAA will expand by orders of magnitude. For instance, a 3200-element ELAA at 2.4 GHz was developed in [2]. With an array size of $2\text{ m} \times 3\text{ m}$, its Rayleigh distance is about 200 meters, which is larger than the radius of a typical 5G cell. Thus, near-field communications will become essential for future 6G mobile networks where the spherical propagation model has to be considered, which is obviously different from the existing far-field 5G systems. Unfortunately, the near-field propagation introduces several new challenges in ELAA systems, which should be identified and addressed to empower 6G communications.

In this article, we systematically investigate near-field communication techniques for 6G. The key features of this article can be summarized as follows:

- The fundamental difference between the far-field and near-field communications is explained. Specifically, the planar wavefront in the far-field can steer the signal energy towards a specific physical angle, while the spherical wavefront in the near-field achieves energy focusing on both angle and distance. Moreover, the Rayleigh distance

that quantifies the boundary between far-field and near-field regions is introduced, and its derivation is explained in detail. Based on this derivation, we further extend the classical Rayleigh distance, for MIMO channels with a direct base station (BS)-user equipment (UE) link, to the one for RIS-aided communications, where a cascaded channel is utilized for presenting the BS-RIS-UE link.

- We investigate the emerging near-field communication techniques by classifying them into two types, i.e., techniques addressing the challenges and those exploiting the potentials in near-field regions. On the one hand, as most techniques specific to far-field often suffer from a severe performance loss in the near-field area, the first type of techniques aims to compensate for this loss, such as near-field channel estimation and beamforming. On the other hand, the second kind of study has revealed that the nature of near-field spherical wavefront can also be exploited to provide new possibilities for capacity enhancement and accessibility improvement. The principles, recent progress, pros and cons of these two categories of research are discussed in detail.
- Several open problems and future research directions for near-field communications are pointed out. For example, the improvement of Rayleigh distance considering various communication metrics need to be analyzed, artificial intelligence (AI) is expected to enable high-performance near-field transmissions with low complexity, and hybrid far- and near-field communications also require in-depth study.

II. FUNDAMENTALS OF NEAR-FIELD COMMUNICATIONS

In this section, we first present the differences between far-field and near-field communications. Then, we will identify the principle to determine the boundary between the far-field and near-field regions in several typical application scenarios.

A. Far-Field Communications vs. Near-Field Communications

The key features of far-field and near-field communications are shown in Fig. 1. We consider an uplink communication scenario, while the discussions in this article are also valid for downlink scenarios. The BS is equipped with an ELAA. A widely adopted metric to determine the boundary between far-field and near-field regions is the Rayleigh distance, also called the Fraunhofer distance [4]. When the communication distance between the BS and UE (BS-UE distance) is larger than the Rayleigh distance, the UE is located in the far-field region of the BS. Then, the EM waves impinging on the BS array can be approximately modeled as planar waves. By contrast, when the BS-UE distance is shorter than the Rayleigh distance, the UE is located in the near-field region of the BS. In this region, the EM waves impinging on the BS array have to be accurately modeled as spherical waves [4].

More specifically, the planar wave is a long-distance approximation of a spherical wave. In far-field regions, the phase of EM waves can be elegantly approximated by a *linear* function of the antenna index through Taylor expansion [4]. This concise linear phase forms a planar wavefront only related to an incident

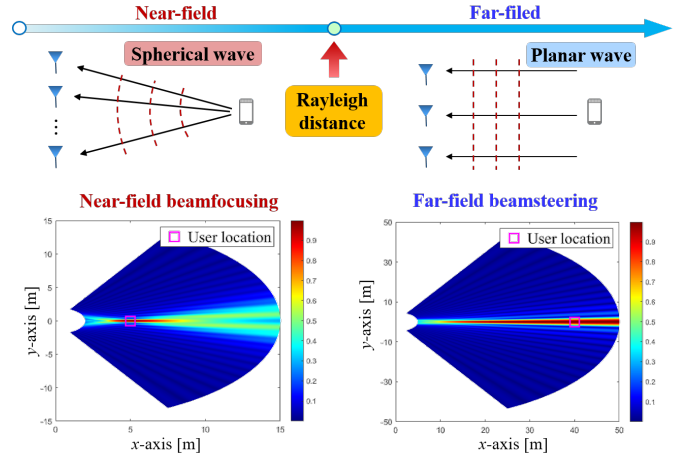


Fig. 1. Far-field planar wavefront vs. near-field spherical wavefront. The bottom two figures illustrate the normalized received signal energy in physical space achieved by near-field beamfocusing (bottom left) and far-field beamsteering (bottom right).

angle. Accordingly, by the utilization of planar wavefronts, far-field beamforming can steer the beam energy towards a specific angle over different distances, which is also termed beamsteering, as shown in the bottom right figure of Fig. 1. Unfortunately, this concise linear phase fails to thoroughly reveal the information of spherical waves. In near-field regions, the phase of spherical waves should be accurately derived based on the physical geometry, which is a *non-linear* function of the antenna index. The information of the incident angle and distance in each path between BS and UE is embedded in this non-linear phase. Exploiting the extra distance information of spherical wavefronts, near-field beamforming is able to focus the beam energy on a specific location, where energy focusing on the distance domain is achievable as shown in the bottom left figure of Fig. 1. Owing to this property, beamforming in the near-field is also called beamfocusing.

The differences between the far-field planar wavefront and near-field spherical wavefront bring several challenges and potentials to wireless communications, which will be detailed in the following sections.

B. Rayleigh Distance

The most crucial premise for near-field communications is to quantify the boundary between the far-field and near-field regions, i.e., the specific value of Rayleigh distance. Generally, the classical Rayleigh distance is proportional to the square of the array aperture and the inverse of the wavelength. Its specific definition can be explained as follows [4]. The true phase of the EM wave impinging on a BS antenna has to be calculated based on the accurate spherical wave model. In the far-field scenario, this phase is generally approximated by its first-order Taylor expansion based on the planar wavefront model. This approximation results in a phase discrepancy. As distance decreases, this phase discrepancy increases. When the largest phase discrepancy among all BS and UE antennas reaches $\pi/8$, the distance between the BS array center and the UE array center is defined as the Rayleigh distance. Furthermore, if the communication distance is shorter than the Rayleigh distance,

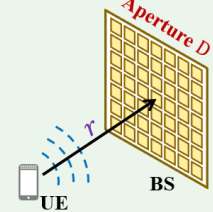
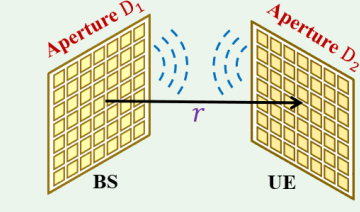
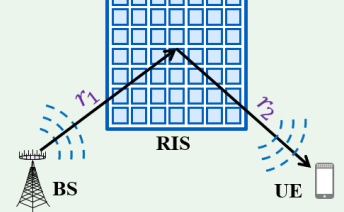
Communication scenarios	SIMO/MISO Wavelength λ Aperture D 	MIMO Wavelength λ Aperture D_1 Aperture D_2 	RIS Wavelength λ Aperture D 
Near-field ranges	$r < \frac{2D^2}{\lambda}$	$r < \frac{2(D_1+D_2)^2}{\lambda}$	$\frac{r_1 r_2}{r_1 + r_2} < \frac{2D^2}{\lambda}$

Fig. 2. Near-field ranges for typical scenarios.

the largest phase discrepancy will be larger than $\pi/8$. In this case, the far-field approximation becomes inaccurate, and thus the near-field propagation needs to be utilized.

Based on the definition of Rayleigh distance, the near-field ranges for SIMO, MISO, and MIMO communication systems can be obtained. As illustrated in Fig. 2, the near-field range of SIMO/MISO scenarios is precisely determined by the classical Rayleigh distance, which is proportional to *the square of BS array aperture*. For the MIMO scenario, since ELAAs are employed at two sides of the BS-UE link, both the BS array aperture and the UE array aperture will contribute to the Rayleigh distance, i.e., the near-field range is proportional to *the square of the sum of BS array aperture and UE array aperture*.

Interestingly enough, in this article, we further extend the conventional Rayleigh distance derived in SIMO/MISO/MIMO systems to that in RIS-aided communication systems, as shown in Fig. 2. Unlike the SIMO/MISO/MIMO channel with a direct BS-UE link, the cascaded BS-RIS-UE channel in RIS systems comprises the BS-RIS and RIS-UE links. Therefore, when calculating phase discrepancy, the BS-RIS distance and the RIS-UE distance need to be added. Then, analog to the derivation of classical Rayleigh distance, the derived near-field range in RIS systems is shown in Fig. 2, which is determined by *the harmonic mean of the BS-RIS distance and the RIS-UE distance*. It can be further implied from Fig. 2 that, as long as any of these two distances is shorter than the Rayleigh distance, RIS-aided communication is operating in the near-field area. Therefore, near-field propagation is more likely to happen in RIS systems.

With the dramatically increased number of antennas and carrier frequency, the near-field range of ELAA considerably expands. For instance, we have recently fabricated a 0.36-meter-aperture ELAA at 28 GHz. If it is employed in SIMO/MISO scenarios, its near-field range is about 25 meters. When both transmitter and receiver are equipped with this array, the near-field range becomes 100 meters or so. Moreover, if this ELAA works as a RIS with a BS-RIS distance of 50 meters, the near-field propagation model should be accepted once the RIS-UE distance is shorter than 50 meters. **In summary, near-field communications come to be an indispensable part for future 6G.**

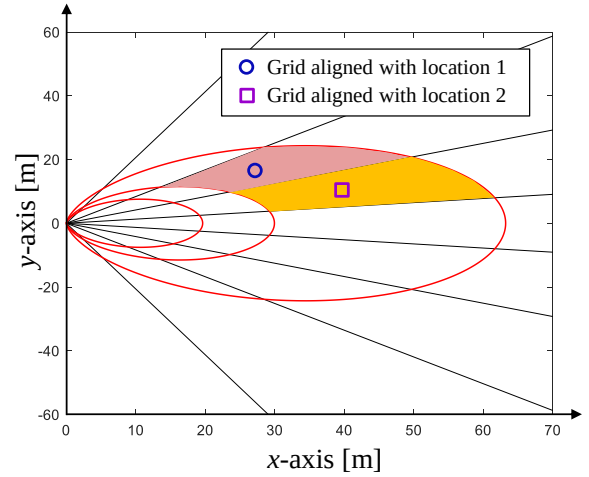


Fig. 3. Near-field codebook with non-uniform grids.

III. CHALLENGES OF NEAR-FIELD COMMUNICATIONS

The near-field propagation causes several challenges to wireless communications, i.e., existing 5G transmission methods specific for far-field suffer from severe performance loss in near-field areas. Technologies recently developed for addressing these challenges are discussed in this section.

A. Near-Field Channel Estimation

Challenge: Accurate channel estimation is required to attain the expected performance gain of ELAA. As the number of channel paths is usually much smaller than that of antennas, channel estimation methods with low pilot overhead generally design suitable codebooks to transform the channel into a sparse representation. For far-field codebooks, each codeword of the codebook corresponds to a planar wave associated with one incident angle. Ideally, each far-field path can be represented by *only one* codeword. With this far-field codebook, the angle-domain representation of the channel can be obtained, and it is usually sparse due to the limited paths. Then, beam training and compressed sensing (CS) methods are applied to estimate far-field channels with low pilot overhead accurately. However, this far-field planar-wave codebook mismatches the actual near-field spherical-wave channel. This mismatch induces that, a single near-field path should be jointly described by *multiple* codewords of the far-field codebook. Accordingly, the near-field angle-domain channel is not sparse anymore, which inevitably leads to the degradation of channel estimation

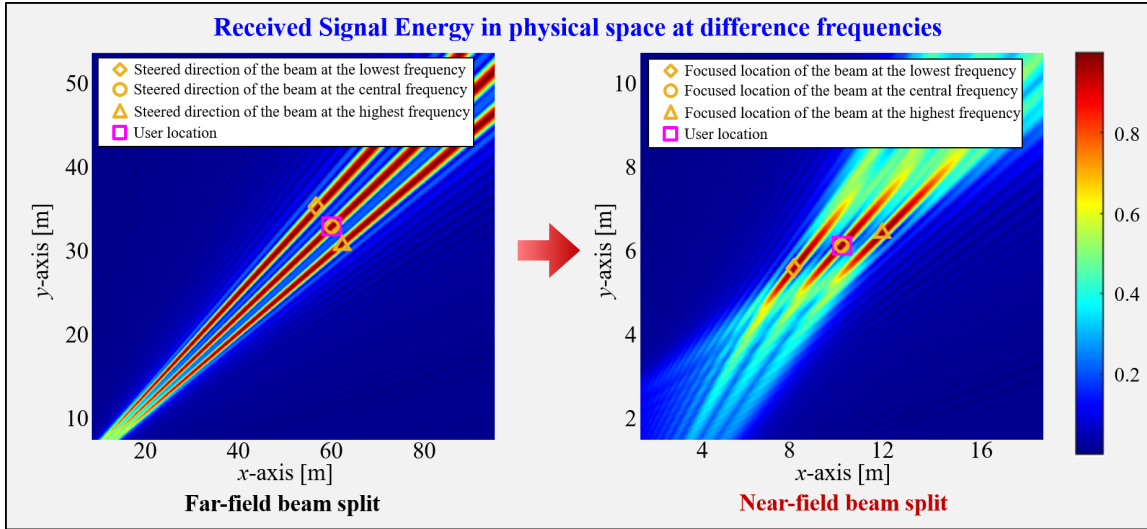


Fig. 4. This figure illustrates the far-field beam split effect (left) and the near-field beam split effect (right). Far-field beam split makes beams at different frequencies transmit towards different directions, while near-field beam split makes beams at different frequencies be focused on various locations.

accuracy. Therefore, near-field codebooks suitable for near-field channels need to be carefully created.

Recent progress: Some recent efforts have been endeavored to design near-field codebooks utilizing spherical wavefronts [5], [6]. In [5], the entire two-dimensional physical space is uniformly divided into multiple grids. Each grid is associated with a near-field array response vector, and all of these vectors construct the near-field codebook. With this codebook, the joint angle-distance information of each near-field path is extracted. Then, the near-field channel can be estimated by CS methods with low pilot overhead. However, with the decrease of BS-UE distance, the near-field propagation becomes dominant, and the distance information gradually becomes more crucial. Therefore, we can conceive the intuition that, the grids should be sparse far away from the ELAA but dense near the ELAA. Without considering this intuition, the codebook in [5] is hard to attain satisfactory channel estimation performance in the entire near-field region. To this end, by minimizing the largest coherence of codewords of the near-field codebook, authors in [6] mathematically prove this intuition, i.e., the angle space could be uniformly divided, while the distance space should be non-uniformly divided. As shown in Fig. 3, the shorter the distance, the denser the grid. With the help of this non-uniform codebook, a polar-domain sparse channel representation and corresponding CS-based algorithms are proposed in [6] to accomplish accurate channel estimation in both near- and far-field areas.

B. Near-Field Beam Split

Challenge: In THz wideband systems, ELAA might encounter a beam split phenomenon, also known as beam squint and spatial-wideband effect. Existing THz beamforming architecture often employs analog phase-shifters (PSs) [7], which usually tune the same phase shift for signals at different frequencies. Nonetheless, the actual phase of the EM wave is the product of the signal propagation delay and the frequency-dependent wavenumber. As a result, the phase of radiated signals can only be well compensated at one frequency by

PS. Phase errors are introduced for the other frequencies, thus causing the beam split effect. In far-field, beam split leads to the fact that beams at different frequencies are aligned with different angles, and the received energy is reduced, as shown in the left figure of Fig. 4. In fact, intensive works have been proposed to mitigate far-field beam split by tuning frequency-dependent phase shifts with planar wavefront through true-time-delay-based (TTD-based) beamforming instead of PS-based beamforming. For near-field beam split, however, beams are focused at both different angles and various distances, as shown in the right figure of Fig. 4. That is because of the split of spherical waves. Owing to the difference between planar and spherical waves, existing schemes addressing the far-field beam split effect cannot work well anymore in the near-field, which poses challenges to THz ELAA communications.

Recent progress: Recently, a few works have tried to overcome the near-field beam split effect. In [8], a variant of chirp sequence is utilized to design the phase shifts, so as to flatten the beamfocusing gain across the frequencies with the sacrifice of the maximum beamfocusing gain. This method can slightly alleviate the near-field beam split effect, but its spectral efficiency degrades as well when the bandwidth is very large, as the beams are still generated by PSs. To this end, a phase-delay focusing (PDF) method is proposed in [9] exploiting TTD-based beamforming. To further illustrate, the BS ELAA is first partitioned into multiple sub-arrays. The UE is located in the far-field area of each small sub-array but within the near-field range of the ELAA. Then, one TTD line is inserted between each sub-array and the radio-frequency (RF) chain to realize frequency-dependent phase shifts. Finally, the frequency-dependent phase variations across different sub-arrays induced by spherical wavefronts are compensated by the inserted TTD line. As a result, beams over the working band are focused at the target UE location [9].

In conclusion, the first solution [8] follows the PS-based beamforming, which is easy to implement but the achievable performance is not satisfactory. The second scheme [9] can nearly eliminate the near-field beam split effect but requires the

implementation of TTD lines. In fact, although deploying TTD lines by optical fibers has been demonstrated in the optical domain, this kind of deployment is non-trivial to be extended to THz ELAA communications. Fortunately, the recent advances in graphene-based plasmonic waveguides provide low-cost solutions for tuning frequency-dependent phase shifts in THz ELAA [7].

IV. POTENTIALS FOR NEAR-FIELD COMMUNICATIONS

Unlike the aforementioned works for dealing with the performance degradation in the near-field, some recent studies have surprisingly revealed that 6G networks can also benefit from the near-field propagation. In this section, we will discuss those studies exploiting the potentials of near-field propagation to improve communication performance.

A. Capacity Enhancement

Potential: The spatial multiplexing gain of MIMO communications considerably increases with the transition from far-field regions to near-field regions. In far-field MIMO communications, the LoS channel can be represented by a rank-one matrix, where the spatial DoFs are very limited. By contrast, the near-field LoS channel can be rank-sufficient derived from the geometric relationship under spherical propagation model. The increased rank indicates a dramatically increased spatial DoFs in the near-field region. Based on the expansion of prolate spheroidal wave functions, it is proved in [10] that near-field spatial DoFs are proportional to the product of the BS and UE array apertures, and inversely proportional to the BS-UE distance. This conclusion is further improved in [11] by meticulously designing the beamfocusing vectors of the BS and UE arrays. As shown in Fig. 5, the DoFs increase from 1 to 20 when the BS-UE distance decreases from 350 meters to 10 meters. Thanks to the increased DoFs, the near-field LoS path enables simultaneous transmission of multiple data streams by MIMO precoding, as opposed to the rank-one far-field LoS channel supporting only one data stream. The increased spatial DoFs can be exploited as an additional spatial multiplexing gain, which offers a new possibility for a significant capacity enhancement.

Recent progress: Recently, some novel precoding architectures have been proposed to leverage these extra near-field DoFs for MIMO capacity enhancement [12], [13]. Firstly, distance-aware precoding (DAP) is developed in [12]. Unlike classical hybrid precoding with a fixed and limited number of RF chains (e.g., 2 or 4 fixed RF chains), the DAP architecture could flexibly adjust the number of RF chains to match the distance-related DoFs, which is achieved by deploying a selection network to configure each RF chain as active or inactive. For instance, in the far-field region, only one RF chain is activated for data transmission. When communication distance is decreasing to 10-20 meters, around 20 activated RF chains are enough to adapt to the DoFs, as shown in Fig. 5. By doing so, the number of transmitted data streams dynamically matches the DoFs. Simulations demonstrate the DAP could significantly increase the spectral efficiency while its energy efficiency is comparable with hybrid precoding. To avoid the utilization

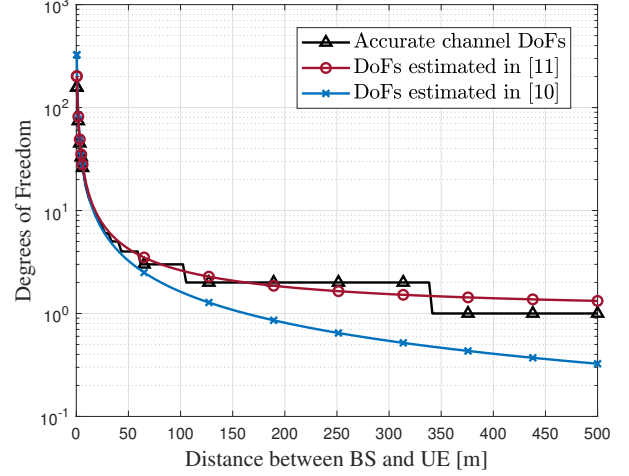


Fig. 5. The spatial DoF increases in the near-field region.

of extra RF chains, another effort to harvest the potential spatial multiplexing gain is the widely-spaced multi-subarray (WSMS) precoding [13]. In this architecture, the sub-arrays are widely spaced to enlarge the array aperture, artificially creating the expansion of the near-field region. Compared with classical hybrid precoding, the number of sub-arrays and the sub-array spacing should be additionally designed in the WSMS architecture. To this end, [13] first assumes planar-wave propagation within each sub-array and spherical-wave propagation across different sub-arrays similar to [9], and then jointly optimizes the number of sub-arrays, their spacing, and the precoding matrix for maximizing the achievable capacity. Simulations demonstrate that WSMS could achieve nearly 200% higher spectral efficiency than classical hybrid precoding.

B. Accessibility Improvement

Potential: Near-field propagation is also able to improve accessibility in multi-user communications. To increase the spectral efficiency in MIMO communications, space division multiple access (SDMA) is widely considered to distinguish different users through orthogonal or near-orthogonal spatial beams. Thus, different users are able to share the same time and frequency resources. For far-field SDMA, far-field beams are generated to align with specific angles based on the planar-wave propagation model. Users located at a similar angle within a spatial beam will severely interfere with each other, and thus can not simultaneously access the network through SDMA. However, as shown in Fig. 6, for the near-field SDMA, near-field beams can be designed according to the spherical wavefront to match specific angles and distances. In this way, users locating at a similar angle but at different distances can simultaneously access the same BS through SDMA.

Recent progress: Taking advantage of these near-field beams, the authors in [14] have studied the near-field multi-user transmission considering the transmissive reconfigurable meta-surface (RMS). By optimizing the sum rate in multi-user systems through alternating optimization, the BS can naturally

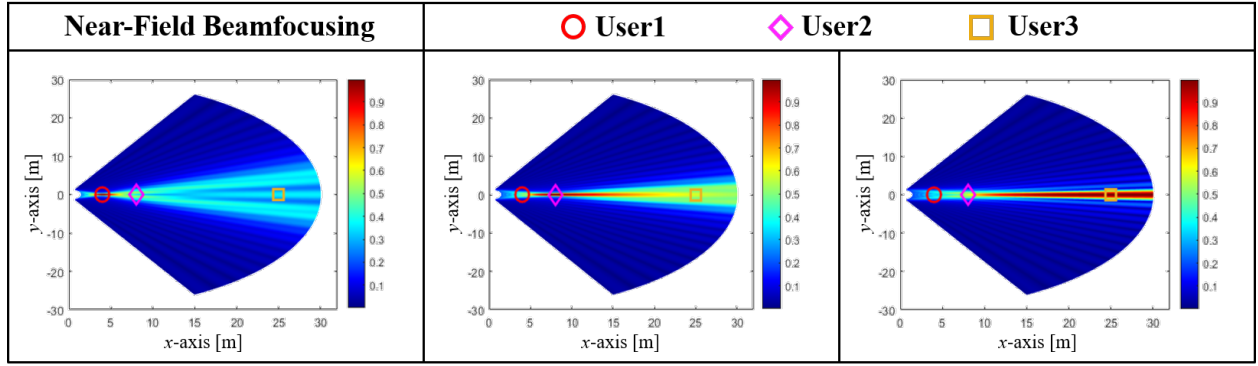


Fig. 6. Near-field beamfocusing is able to serve multiple users in the same angle.

generate focused beams with spherical wavefronts to distinguish users located at a similar angle but at different distances. The results demonstrate that near-field propagation has the potential of enhancing multi-user accessibility.

V. FUTURE RESEARCH DIRECTIONS

In this section, several future research directions for near-field communications will be pointed out.

A. Near-Field Communication Theory

1) *Improvement of Rayleigh Distance*: As a widely adopted quantification of near-field range, Rayleigh distance is attained in terms of phase discrepancy. For communication metrics directly affected by phase discrepancy, such as channel estimation accuracy, Rayleigh distance can accurately capture the performance loss of applying far-field channel estimation schemes in the near-field region. On the contrary, some metrics are directly influenced by other factors instead of phase discrepancy, e.g., capacity is determined by beamforming gain and channel rank. Accordingly, classical Rayleigh distance probably cannot capture the performance loss of these metrics well. To this end, several recent works have endeavored to improve classical Rayleigh distance in terms of some vital communication metrics. For instance, an effective Rayleigh distance (ERD) is derived in [9] for the accurate description of beamforming gain loss and capacity loss. Nevertheless, ERD is only valid for MISO communications, while more discussion should be made to improve Rayleigh distance in more practical scenarios under more general metrics, e.g., channel rank and energy efficiency.

B. Near-Field Transmission Technologies

1) *AI-Aided Near-Field Communications*: Different from far-field communications, the transmission algorithms for near-field communications are more complex. To be specific, since extra grids on the distance domain are required as mentioned in Section III-A [6], the size of near-field codebooks is usually much larger than that of far-field codebooks, leading to high-complexity channel estimation and beamforming. Moreover, the non-linear phase characteristics of spherical waves make the design of near-field beam training algorithms more complicated

than that in far-field areas. AI-based transmission methods are promising to address these problems since they can mine the features of near-field environments through *non-linear* neural networks. Currently, there are plenty of works elaborating on AI-based far-field transmissions (references are not provided here since they are limited in this magazine), while the AI-based near-field transmissions have not been well studied.

2) *RIS-Aided Near-Field Communications*: Compared with MIMO communications, the near-field propagation becomes even more dominant and complex in RIS-aided systems. In MIMO communications, based on the spherical propagation model, the EM waves form *spherical* equiphase surfaces at the receiver. On the other hand, in RIS-aided systems, the phase of received EM waves is accumulated by the propagation delays through the BE-RIS and RIS-UE links. Based on the geometry relationship, the equiphase surfaces become *ellipses* in the near-field range instead of spherical. Accordingly, the research on beamfocusing [15], channel estimation, and multiple access techniques taking into account this ellipses-equiphase property are required for RIS-aided near-field communications.

3) *Hybrid Far- and Near-Field Communications*: In practical systems, communication environments usually exist with both far-field and near-field signal components. First, in multi-user systems with multi-path channels, some users and scatterers may be far away from the BS while others are located in the near-field region of the BS, which constitutes a hybrid far- and near-field (hybrid-field) communication scenario. Additionally, it is worth mentioning that the Rayleigh distance is proportional to frequency. Thus, in an ultra-wideband or frequency-hopping system with a very large frequency span, its near-field range varies dramatically across the bandwidth. Chances are that the signal components at low frequencies may operate in far-field regions while those at high frequencies with larger Rayleigh distances are propagating in the near-field areas, which also contributes to hybrid-field communications. Consequently, the above factors make hybrid-field communications practical and crucial in future 6G networks. Thus, hybrid-field transmission techniques handling both far-field and near-field signal components deserve in-depth study.

4) *Spatial Non-Stationarity Effect on Near-Field Communications*: Except for near-field propagation, the spatial non-stationarity effect is another fundamental characteristic of ELAA compared to 5G massive MIMO, where different

scatterers and users are visible to different portions of the ELAA. This effect leads to the fact that only a part of the ELAA can receive the spherical EM waves radiated by a scatterer or a user. The angular power spectral and average received power rapidly vary over the ELAA. Recently, there have been intensive works dealing with the non-stationarity effect and near-field propagation simultaneously [5]. However, the impact of non-stationarity on other emerging near-field communications has not been well studied, such as RIS-aided systems and hybrid-field communications.

C. Hardware Development

To verify the effectiveness of near-field transmission technologies, hardware developments and over-the-air experiments are of great significance. For example, for alleviating the near-field beam split effect, TTD lines need to be meticulously designed in THz domain. The hardware developments of WSMS and DAP architectures are worth being carried out to exploit the near-field spatial DoFs. Besides, implementing these techniques still has to overcome several hardware impairment issues, including In-phase/Quadrature imbalance, low-efficiency power amplifier at high frequency, etc. All these challenges should be carefully addressed to enable the implementation of 6G near-field communications.

VI. CONCLUSIONS

With the evolution from massive MIMO to ELAA, near-field propagation with spherical wavefront comes to be essential in 6G networks, where conventional far-field propagation with planar wavefront is not valid anymore. In this article, we reveal that near-field propagation is a double-edged sword, i.e., it brings both challenges and potentials to 6G communications. We first introduced the non-linear phase property of spherical waves and explained the derivation of near-field range in terms of phase discrepancy. Then, we discussed the technical challenges of channel estimation and beam split caused by near-field propagation, and presented the recent solutions. In addition, some appealing works that exploit the capability of spherical waves to potentially improve spatial DoFs and accessibility are investigated. Several future research directions for near-field communications were also highlighted, which are expected to inspire more innovations on 6G near-field communications.

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